

# LASER-RAMAN SPECTROSCOPY OF LIVING CELLS

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### Abstract:

Investigations into the laser-Raman shift spectra of bacterial and mammalian cells have revealed that many Raman lines observed at 4–6 K, do not appear in the spectra of cells held at 300 K. At 300 K, Raman activity, at set frequencies, is observed only when the cells are metabolically active; however, the actual live cell spectrum, between 0 and  $3400\text{ cm}^{-1}$ , has been found to alter in a specific way with time as the cells' progress through their life cycles. Lines above  $300\text{ cm}^{-1}$ , from *in vivo* Raman active states, appear to shift to higher wave numbers whereas those below  $300\text{ cm}^{-1}$  seem to shift to lower ones. The transient nature of many shift lines observed and the intensity of them when present in the spectrum indicates that, *in vivo*, a metabolically induced condensation of closely related states occurs at a set time in the life of a living cell. In addition, the calculated ratio between the intensities of Stokes and anti-Stokes lines observed suggests that the metabolically induced "collective" Raman active states are produced, *in vivo*, by non thermal means. It appears, therefore, that the energetics of the well established cell "time clock" may be studied by laser-Raman spectroscopy; moreover, Raman spectroscopy may yield a new type of information regarding the physics of such biological phenomena as nutrition, virus infection and oncogenesis.

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## Introduction

Over the past four decades investigations into the working of living cells designed to understand those special properties of matter that constitute life, have been dominated, almost exclusively, by biochemically oriented considerations. Such works, by a large number of dedicated researchers, have (a), increased our knowledge of the chemical composition of life forms; (b), unravelled the chemical structures of some large macromolecules; (c), tracked the pathways by which many components are synthesized and (d), led to the introduction of a mechanism by which the primary structure of proteins *in vivo* may be determined and the required codes handed down from one generation to the next. The sheer number of such investigations and the wealth of information they have provided, however, have tended to overshadow other types of approaches to the problem of life, and to drown the words of caution, issued from time to time, from the voices of those who have walked different paths. It is understandable, therefore, that initial ideas and theory regarding the nature of life were based on chemical concepts and the random collision kinetics of physical chemistry.

In more recent times cell biologists and geneticists have uncovered new facts which reveal the cell to be a highly ordered, dynamic entity in which the chemical events, described by the biochemists, occur in sequences of extreme precision with respect to time, place and accuracy; few mistakes, if any, are made in either of these three parameters. In addition the rates, at which the *in vivo* syntheses of macromolecules occur are several orders of magnitude faster than those performed, in the test tube, by individual components in isolation from the rest of the cell. From the standpoint of statistics and the enormous number of steps performed by a cell, in just one sequence of events, the probability of an error, if it was governed by random collision, would far exceed the small number of errors actually made. Some process exists *in vivo*, therefore, able to direct, in time and space, rapid sequential events with great accuracy, a process accredited by biologists, to the "cell time clock"; giving the phenomenon a name, however, does not provide, in any way, a clue to its mechanism.

The living cell has been found to possess unexpected electrical properties, with membrane potentials  $10^7$  V/M and dielectric characteristics which alter with time and nutrition, phenomena which have occupied the research efforts of some scientists over many years. It is susceptible also to injury from specific frequencies of electromagnetic radiations of extremely low photon energy, such as microwaves, by mechanisms which appear to be of a non-thermal nature [1, 2, 3, 4]. While the cell is stable and capable of extreme accuracy in its internal manoeuvres, under "normal" thermal conditions, its precision can be seriously impaired by an input of small quantities of electromagnetic energy. The livelihood of the cell thus rests heavily on the specific, yet delicately balanced, physical relationship it creates between its component parts; balances which it is able to generate and maintain sometimes in the presence of excess thermal energy.

The creation of specific states and transitions between them are known vital parts of the life process. Somehow, however, the cell is able to direct, in time and space, the occurrence of these transitions so as to extract configurational energy from its nutrients, with a minimum of loss in the form of heat or the emission of electromagnetic radiation, and produce an entity identical to itself, or one with a limited lifetime but special mission within societies of cells that make up multicellular life forms.

It is clear today that life, even in its simplest form, the cell, does not rest on the independent activities of enzymes, DNA or even membrane potentials but arises from the creation of unique complexes of macromolecules in which the chemical and physical characteristics of the individual molecules are changed and their activities integrated. It is, therefore, a collective phenomenon of enormous complexity as far as the structure and activities of its component parts are concerned, but one

which may be understood more easily perhaps, through the acquisition of knowledge regarding the physics of the phenomenon, as Fröhlich has attempted to do over the years [5, 6]. Much of the enormous effort of biochemists and cell biologists could be wasted, therefore, if physicists do not take up the challenge of ascertaining how the information supplied by these scientists may be collectively viewed in terms of energy, space, time and motion. Any physicist who does try to solve anyone of these problems, however, must keep in mind the limitations of the theories of physics based on “simple” models in which he or she has been schooled. Most, if not all of these amply apply to an individual atom or small collections of them or to collections of identical but small molecules; there is, however, no guarantee that these theories will hold true as far as the life phenomenon is concerned. Indeed, one of the excitements of such explorations is the possibility today of new physical “laws”, or phenomena, being discovered via studies of the life process just as they were a century or so ago. Moreover, while it is important that the physicist acquaints himself with the fundamentals and theory of modern biochemistry and cell biology (which may seem, at first, a formidable task, although the essentials, free of intimate details, are not difficult to grasp) it is equally important that he appreciates information acquired much earlier, something which even modern biologists often fail to do. It is essential to recognize, for instance, that ever since chromosomes and the developing embryo were first viewed under a primitive microscope directed and orderly motions of macromolecular complexes and whole cells have been known to be vital parts of the life process; these activities have been given names, such as mitosis and meiosis etc., but the mechanism by which they occur remains neither explained nor covered by popular biological theory. Another feature of the cell is that simply to survive as a life form it requires no use of outside energy, inputs of thermal or electromagnetic energy can be “anti-life” but if these are avoided a cell can retain, for an indefinite period, its ability to renew its life activities if the correct environmental conditions are established. In this state of anabiosis, a condition not understood, the cell possesses, in the form of enzymes, electrical potentials, and so on, all the energy necessary to perform its tasks. It lacks only the ingredients with which to do them. If these are supplied, it simply uses its inherent energy to re-arrange the configurational energy, present in the supplied nutrients, to form a unit just like itself, or to make something other cells require. No one can say how self-perpetuating systems, able to re-arrange random energy sources into identical organized systems, evolved, all that can be said today is that they exist and constitute the only method by which energy “packets” with the property of life may be reproduced. Since all the components of “ruptured” cells in a test tube will not reform into cells, it may be said safely that it takes life to create life and it is with life that we must work if we are to understand it.

In the knowledge that the cell is a special form of integrated changes in energy states transformed by metabolic processes into coherent and specific vibrations of large macromolecular complexes and mechanical motions (analogous to the crystals and cogs of a highly accurate watch) some researchers have tried to bring to bear, on the problem of understanding the energy transactions of life, new physical tools such as the laser and Raman spectroscopy. The purpose of this report, therefore, is to describe the difficulties and frustrations encountered in such endeavours and the preliminary successes achieved.

### **Why laser-Raman spectroscopy?**

If life is to be considered a collective phenomenon and to arise from a particular way in which transitions in the states of large macromolecules are integrated then, to understand the life “force”, methods must be used which will display not only the energy distribution in isolated molecules but also

how this is affected by the formation of macromolecular complexes and the presence of adjacent ones. In addition, the techniques should permit a determination of those transitions allowed and propagated *in vivo* by living entities. Some form of spectroscopy, therefore, is suggested. Many powerful tools exist such as Nuclear Magnetic (NMR) and Electron Spin Resonance (ESR) but these require, often, the addition to the system of special chemicals toxic to the living cell. The same is true of methods which employ gamma rays, X-rays and ultraviolet light, all have been, and are, invaluable techniques for molecular biology but all are lethal to life processes. To look at life the best techniques are those which do the least harm. Thus visible and infrared light appear to be the preferred probes.

Straight forward absorption spectroscopy in the visible would seem to be insufficient. Only a few biological molecules absorb in the visible and those that do usually have very specialized functions, in vision, photosynthesis and so on, and while vision is a part of some life forms it is not essential to life as such. Photosynthesis in the overall picture is essential to life on earth but it plays no direct role in the organization of animal and other heterotrophic life forms. Thus while valuable information about the above phenomenon has been obtained by such absorption techniques, the procedures have limitations which may prevent them from yielding the type of knowledge sought.

Since it is the allowed transitions in states *in vivo* about which information seems vital, infrared or Raman spectroscopy appear the methods of choice. In accordance with classical electrodynamics any motion of an atomic system involving a change in dipole moment leads to absorption, or emission, of radiation. Any vibrational motion of a molecule may be resolved into a sum of normal vibrations and, as these are simple periodic motions, the frequencies are the ones emitted or absorbed and lie in the near infrared. To be infrared active, the vibrations must be connected with a change in the dipole moment ( $M$ ), so in the harmonic oscillation approximation only the fundamentals are active. In asymmetric molecules, however, every normal vibration is infrared active, only in symmetrical molecules may the change in dipole moment be zero and thus a motion be infrared inactive.

In order for a fundamental to appear as a Raman shift, the amplitude of  $M$  must change. The magnitude of the dipole moment  $p = \alpha E$ , where  $E$  is the electric vector of the incident radiation of frequency  $\nu$  and  $\alpha$  = polarizability. If, therefore,  $\alpha$  changes during a vibration with a frequency  $\nu_1$ ,  $P$  will change with frequencies  $\nu + \nu_1$ ,  $\nu - \nu_1$  and  $\nu$ . In asymmetrical molecules, all normal vibrations are associated with a periodic change in polarizability and are Raman active. From classical considerations an induced dipole moment will oscillate at frequency  $\nu$  of the incident radiation (Rayleigh scattering) and at  $\nu \pm \nu_1$  (Raman scattering); the vibration  $\nu_1$  will appear as a Raman shift, however, only if one of the components of the polarizability  $\alpha$ , for the various scattering geometries, differs from zero. From classical theory, therefore, the intensity of the Raman line  $\nu - \nu_1$  should equal that of the  $\nu + \nu_1$ .

From quantum theory the frequencies of infrared absorption and Raman shift bands are determined by the energy difference between the levels of a transition with the selection rule  $\Delta\nu = \pm 1$  for each normal vibration. For molecules with a centre of symmetry infrared active transitions can occur only between states of opposite symmetry with respect to the centre of symmetry, i.e. components of  $M$  are antisymmetric but Raman active ones can occur only between states of the same symmetry. For all molecules, however, without a centre of symmetry many transitions can be both infrared and Raman active. As far as sublevels are concerned, only those of opposite parity are IR active whereas those of the same parity are Raman active. Rotational transitions follow similar selection rules in the IR, a transition can take place only if there is a permanent dipole moment and positive levels can combine only with negative ones, thus  $\Delta J = \pm 1$ ; for Raman active transitions the selection rule is  $+\leftrightarrow +$  and  $-\leftrightarrow -$  and  $\Delta J = 0 \pm 2$ .

From the above it is clear that the wave number of IR and Raman shift bands are equal to the actual

frequencies of the movements, thus from classical and quantum theory the spectra are the same as far as position and occurrence of bands are concerned. There is, however, one important difference, classical theory suggests that the intensity of Raman bands  $\nu - \nu_1$  (Stokes lines) and  $\nu + \nu_1$  (anti-Stokes lines) are the same, whereas quantum theory predicts that the number of vibrations in the first state of the anti-Stokes line is only  $e^{-h\nu/kT}$  times the number of vibrations in the initial state of the Stokes line (e.g. the ground state). Thus anti-Stokes lines are considerably weaker than their Stokes counterpart and are rarely seen at wave numbers above  $500\text{ cm}^{-1}$ . It is this difference, therefore, that makes Raman spectroscopy more attractive for biological endeavours than infrared.

The large molecules of life systems are, in the main, linear and asymmetrical thus both IR and Raman active. The metabolically induced electronic transitions between them, *in vivo*, moreover, should involve not only changes in dipole moment but in large alterations in their amplitude, and hence polarizability; and, since it is controlled shifts in electron density which seem to control life process, such processes should exhibit Raman activity. A living system, however, it not governed by thermal equilibria as, in order for it to function, it must produce "excited" states at what must be considered to be low temperatures. An amino acid, for instance, may be produced by the chemist on refluxing certain chemicals at temperatures lethal, on microsecond exposures, to all life forms. The cell, however, produces them, at rapid rates, at very low temperatures and it does so by using the "configurational" energy present in its large organic catalysts, namely its enzymes, in place of heat. In general, heat is toxic to life forms and is a waste product of some of their internal processes. The association of even a small molecule, such as  $\text{H}_2\text{O}$ , with a macromolecule can induce enormous shifts in electron cloud densities causing severe alterations in the conformation of the macromolecule. For instance, transitions between the B and A forms of DNA are brought about by the association or disassociation of just a few molecules of  $\text{H}_2\text{O}$  to or from its outer groups [7, 8, 9]. Thus a rhythmic oscillation between the two conformations can be set up physiologically by  $\text{H}_2\text{O}$  exchange with large changes in dipole moments and their amplitudes. For a cell to "live", it has to create unstable states; a cell in thermal equilibrium, therefore, is either dormant, i.e. in an anabiotic state, or dead. The physical mechanism by which life forms perform their activities suggests, therefore, that large rhythmic shifts in the amplitudes of dipole moments of specific frequencies occur which should give rise to Raman shifts of high intensity, moreover, the population of upper states should differ from that arising from normal thermal distribution. Thus, if this is fact, then the ratio between the intensities of Stokes and anti-Stokes lines should differ from the thermal factor and anti-Stokes lines observed at higher Raman shift frequencies. Raman spectroscopy, therefore, appears to offer a greater versatility in the interpretation of the physics of life processes than infrared, although, for a full picture to be seen, both should be performed.

## Experimental findings

With the advancements in laser technology laser-Raman spectroscopy has proven a powerful tool for studies in many areas of physics, chemistry and molecular biology, by which new information regarding the structure of small and large molecules has been obtained. The excellent works of experimentalists such as Yu and colleagues on the proteins of the eye [10, 11]; Hartmann [12] and Lord [13] on many proteins and nucleic acids have revealed the spectra of these molecules to be complex, reproducible and specific. Data on the spectra of isolated biomolecules [14–20] and membranes [21–25] now are abundant, but it is not, however, the purpose of this report to describe such information and interested readers are referred to the recent reviews of the subject [14–25]. It was the success of these endeavours,

nevertheless, that led the author to believe that the technique should yield new information about *in-vivo* energy states even though the spectra of whole cells, by expectation, would be extremely complex; work on the subject began, therefore, in our laboratories in 1971.

## Stokes spectra

The biological models chosen initially were the microbes *Bacillus megaterium* and *Escherichia coli*, as much is known about their physiology and molecular composition. Following the procedures used by prior investigators with isolated macromolecules the microbial cells were collected and thoroughly washed in 0.85% NaCl (saline) or 0.1 M phosphate buffer (pH 6.9) (Buffer) and then used either as suspensions in saline, Buffer or H<sub>2</sub>O or as a wet "paste". The physical equipment consisted of a high resolution Cary 82 spectrometer, a Spectra-Physics argon-ion laser and a PDB 11 minicomputer with a link to an IBM 370. The spectrophotometer possessed a multipass system, by which some ten passes of the laser beam could be focussed on a spot in the sample, and a sample chamber equipped with facilities by which the sample could be maintained at any desired temperature. The available apparatus, therefore, was of a high quality and conducive to immediate success, but this was not the case. For almost a year my colleagues and I worked with "pastes" and cell suspensions varying from 10<sup>3</sup> to 10<sup>9</sup> cells/ml but without success, the only Stokes spectrum obtained, if one could call it a spectrum, was that of water and the suspending medium which, because of its dilute nature produced only a few weak lines (fig. 1). These results were confusing, therefore, and disappointing as, from the standpoint of physics, the spectra seen, or lack of spectra seen, made little sense. The enterprise was given up, therefore, because of the belief that something was wrong with either the equipment or the techniques employed. Repeated publications in the literature of the spectra of large biomolecules prompted, however, a return to the project but again with no success.

It was our custom to discuss our failures each morning over tea and on such a morning, with the usual "Raman" run in operation, we heard a loud whistle from our recorder, a characteristic of it, even though new, when it recorded a strong line. Over the next few minutes it continued to whistle and produce a spectrum the like of which we had not seen before. A hasty reset of the instrument produced a second and a third spectrum then, disappointment, on the fourth time of asking, no lines appeared. Over the next week total failures again were experienced, but, in checking over all procedures, suspending fluids etc. it was discovered, by Mrs. Montgomery, our technician, that on the "successful" morning she had inadvertently placed the cells in a fluid containing glucose; the salts fluid used, therefore, was not a holding one as thought, but one in which the cells were able to grow. This experiment using a growth medium, therefore, was repeated and, up to a point, it was successful.

The cuvettes used were of a special kind in that, for use with the multipass system, set Brewster angles were required and only a 0.5 ml volume of sample could be employed. With any one sample in the cuvette, only two or three spectra with lines could be obtained, after this the recorded spectra became, once again, essentially devoid of lines. In addition, in the two or three runs able to be made and showing lines, the spectra were almost totally different; if aliquots of fresh cells, however, were placed in the cuvettes, renewed Raman activity was seen but again each spectrum obtained differed, in large part, from that of the others. Later experiments revealed that more spectra with lines could be achieved, with a given sample, if the laser power was reduced (but here the lines were much weaker) and also if a given scan of up to 3400 cm<sup>-1</sup> was conducted as short consecutive scans of 400 cm<sup>-1</sup> with a 2 minute, dark period, between them. Thus the maximum time the cells could be in a 300 mW laser beam and a spectrum produced was

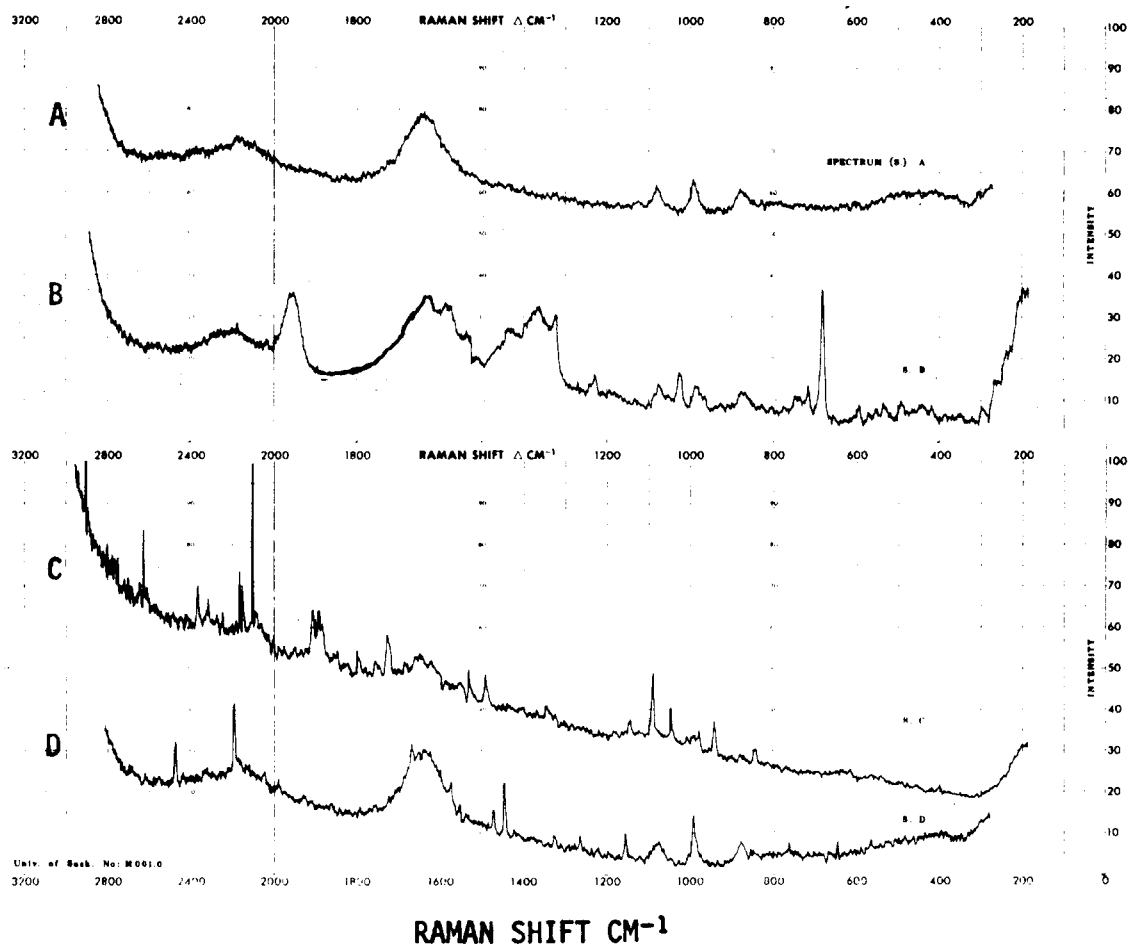


Fig. 1. Examples of initial spectra obtained with washed *B. megaterium* cells in Davis  $\text{NH}_4^+$  salts medium. A, without glucose at any incubation time; B, C and D, with glucose at 30, 60 and 90 minutes of incubation respectively. Sensitivity  $1 \times 10^4$  cps. Band width  $5 \text{ cm}^{-1}$ ; laser 488 nm. Taken from Webb et al. [26].

about 6 minutes. The technique adopted, therefore, was to scan for 3 minutes only and then change to a fresh sample of cells taken from a culture actively growing in a growth medium.

Normal bacteriological growth media, such as the various nutrient broths could not be used because all are highly coloured and hence, in a laser light, they display complex fluorescences. The growth medium of choice, therefore, was the Davis minimal salts medium (MM) which contains only inorganic salts and glucose; if this was used as a 1:3 dilution in  $\text{H}_2\text{O}$  of its normal strength it produced a characteristic Raman shift spectrum with three weak lines at  $873 \text{ cm}^{-1}$ ,  $988 \text{ cm}^{-1}$  and  $1068 \text{ cm}^{-1}$  and one stronger peak at  $1660 \text{ cm}^{-1}$  which was due, in part to water (fig. 1). For cells unable to proliferate in this medium a second one was used which consisted of the MM to which a given amino acid, or group of amino acids, at  $1.0 \mu\text{g/ml}$  was added. At this low concentration no Raman shift lines were produced from these additives and no fluorescence problems were encountered.

## Stokes spectra with synchronized populations

Even with the sequential 3 minute scans and the above growth media the spectrum, from 5 to  $3200\text{ cm}^{-1}$ , varied from run to run (fig. 1) and, moreover, if the temperature in the sample chamber of the spectrophotometer was held between  $15^{\circ}\text{C}$  and  $25^{\circ}\text{C}$  no lines, except those of the medium were observed. It was clear, by now, that, to obtain Raman shift lines from intact cells, the cells had to be in an active metabolic state which required that they be given a usable carbon (energy) source and environmental temperatures above  $25^{\circ}\text{C}$ . Thus, it was concluded that, if these conditions held, the Raman shift lines observed were due to energy states produced *in vivo* by an active metabolism, and the non-reproducibility of the spectrum was because these states changed, with time, during the lifetime of a given cell. To test this hypothesis synchronized populations of cells next were employed.

Synchronization was achieved by taking cells from a stationary phase culture and first; washing them three times with 0.85% NaCl (saline); second, resuspending them at  $10^7$  cells/ml in saline and subjecting them to two successive heat and cold shocks, at  $40^{\circ}\text{C}$  and  $4^{\circ}\text{C}$ , for 15 minutes at each temperature; and third, resuspending them at  $2 \times 10^7$  cells/ml in a fresh medium held at  $35^{\circ}\text{C}$  or  $37^{\circ}\text{C}$ . A second technique was the starvation for 30 minutes, of an auxotrophic strain of *Escherichia coli* K12 of a vital nutrient, such as thymine, and then add the nutrient to the medium. Both techniques produced a population of cells in which the metabolic events in each cell were synchronized, to within 3 to 5 minutes, in approximately 70% of the population. After resuspending the synchronized cells in the new medium, new samples were taken every 3 minutes and the shift spectrum between the Rayleigh line and about  $3200\text{ cm}^{-1}$  determined. This procedure was carried out only over the first generation (or lag phase) of the cells, which varied from 70 to 180 minutes depending on the growth medium used, at a scan rate of  $2.5\text{ cm}^{-1}/\text{sec}$  as this gave the best results. Thus a complete spectrum could be obtained every 20 minutes or repeat segments of  $450\text{ cm}^{-1}$  run every 3 minutes. For the examination and presentation of the results the abbreviation, MAR, was coined to signify the number of minutes after the cells had been resuspended, in the appropriate growth medium, and at which, a spectral run was started.

The four spectra, obtained over a 90 minute growth period, always differed from one another, only two very broad lines between  $1600\text{--}1800\text{ cm}^{-1}$  and  $2000\text{--}2150\text{ cm}^{-1}$  appeared in all but, even here, the fine characteristics of each of these changed (fig. 2). This type of result had been expected, what was not anticipated, however, was the subsequent inability to reproduce, in toto, each of the four spectra in repeat runs: some lines occurred again, others did not and new ones appeared. The reason for this was found to be the inability to reproduce exactly the same characteristics in successive microbial cultures. The lag phase of the cells, after the treatments given, varied by  $\pm 5$  minutes and apparently this variation, which could not be improved upon, was sufficient to alter the spectral characteristics.

At this time much data had been collected and so some 62 complete spectra were examined; each distinct line seen was tabulated, broad lines of more than  $100\text{ cm}^{-1}$  were omitted unless a clear peak appeared on them, as seen in fig. 2A and C at  $1225\text{ cm}^{-1}$  or  $1620\text{ cm}^{-1}$ . In addition the very narrow "spikes", seen in fig. 1C at  $2132\text{ cm}^{-1}$ , were omitted for reasons to be discussed below. When these lines were graphed as a histogram, in terms of the number of times a given line ( $\pm 2.5\text{ cm}^{-1}$ ) was seen, it became clear that the appearance of lines was not random, all the lines occurred over and over again although they were not always present in a given spectrum (fig. 3). The composite, shown in fig. 3, therefore, may be considered as the, in toto, Raman Shift Stokes spectrum of metabolically active *Escherichia coli*. When these lines were histogrammed in terms of the time of their spectral appearance during the life span of the cells they became even more informative. As can be seen from fig. 4 the bulk

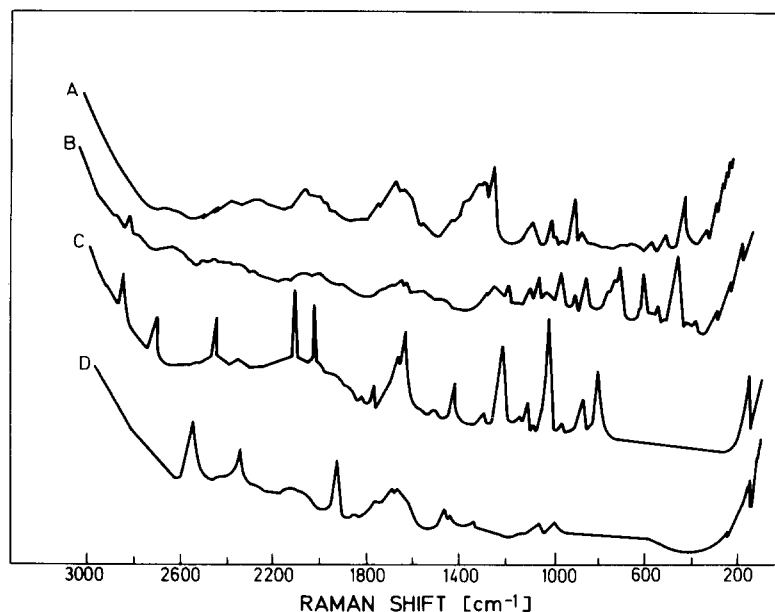


Fig. 2. Examples of spectra obtained with synchronized *Escherichia coli* cells in Davis  $\text{NH}_4^+$  salts. A, 0–20 min; B, 20–40 min; C, 40–60 min; D, 65–85 min. Conditions as for fig. 1.

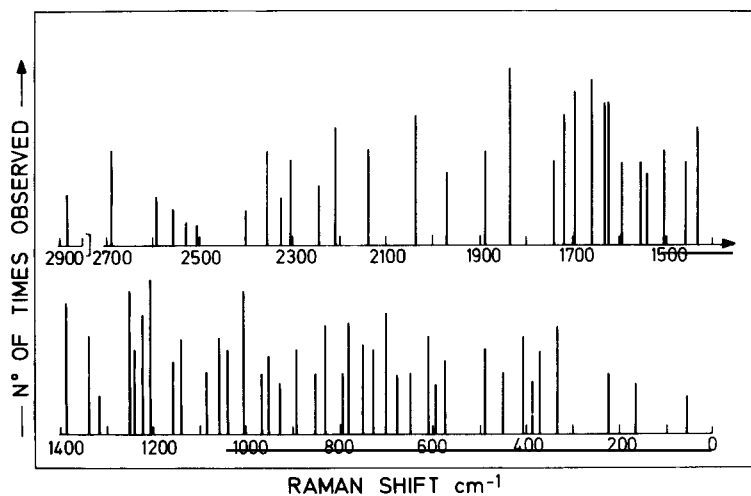


Fig. 3. Composite of all Raman lines observed in the spectra obtained with synchronized populations of *Escherichia coli* cells. Each Raman line seen in a given spectrum, recorded at a set time in the lifetime of the cell, is tabulated along with the number of times each line re-occurred in repeat spectral determinations. The height of the frequency markers, used to scale, the illustration, = two appearances of a line.

of the lines, during the first 30 minutes of the cells' lifetime, occurred between  $300\text{ cm}^{-1}$  and  $1200\text{ cm}^{-1}$ , from 30 to 60 minutes it was between  $800\text{ cm}^{-1}$  and  $1900\text{ cm}^{-1}$  and from 60 to 80 minutes between  $1400\text{ cm}^{-1}$  and  $2600\text{ cm}^{-1}$ . Thus the lines, associated with Raman active metabolic states appeared to shift, with incubation time, to higher wavenumbers; there was, however, an exception to this, the lines below  $200\text{ cm}^{-1}$  appeared to move towards lower wavenumbers (both facts apparent from figs. 1, 2, 3, 14). Another interesting finding was that, approximately 10 minutes before cell divisions began, lines at about  $2035\text{ cm}^{-1}$  and  $2142\text{ cm}^{-1}$  often became very strong (fig. 5).

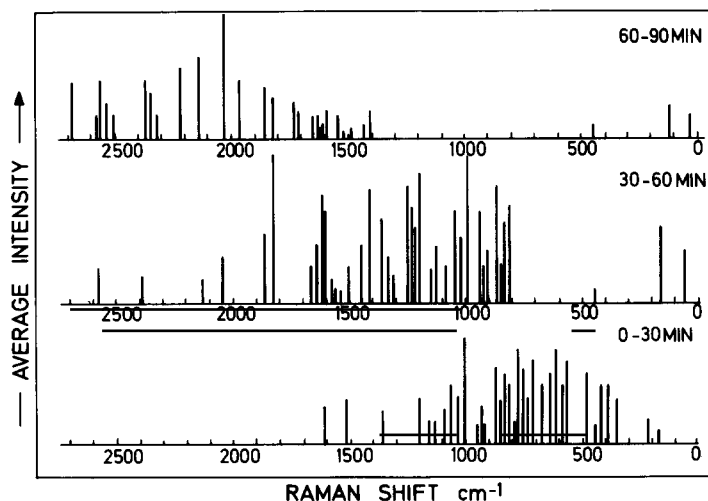


Fig. 4. The timed appearance of Raman lines seen in the spectra of synchronized populations of *Escherichia coli* cells during a 90 minute cell cycle. The times shown represent the time after the cells were placed in a minimal medium and, at which, their Raman shift spectrum was determined. The average intensity of each line is that calculated from two, or more, spectra in which the line appeared and is presented merely to demonstrate the relative intensities of the lines and change in them with time.

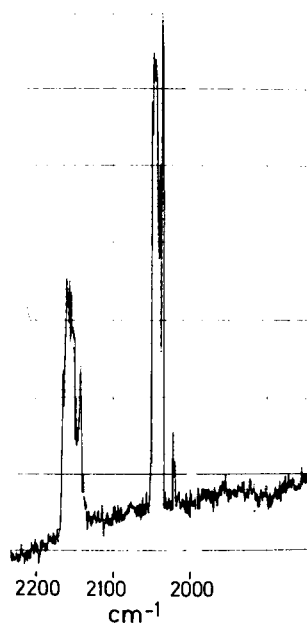


Fig. 5. The strong Raman lines which appeared in the spectrum of *Escherichia coli* just before cell divisions began. Laser 5145 Å, sensitivity 10 K.

### Low temperature spectra

The spectra of *Escherichia coli* at 6 K recently obtained is shown in fig. 6. These spectra were produced from a 20 mg, wet weight, paste of 18 hr cells, grown in MM at 33°C, placed on a silvered

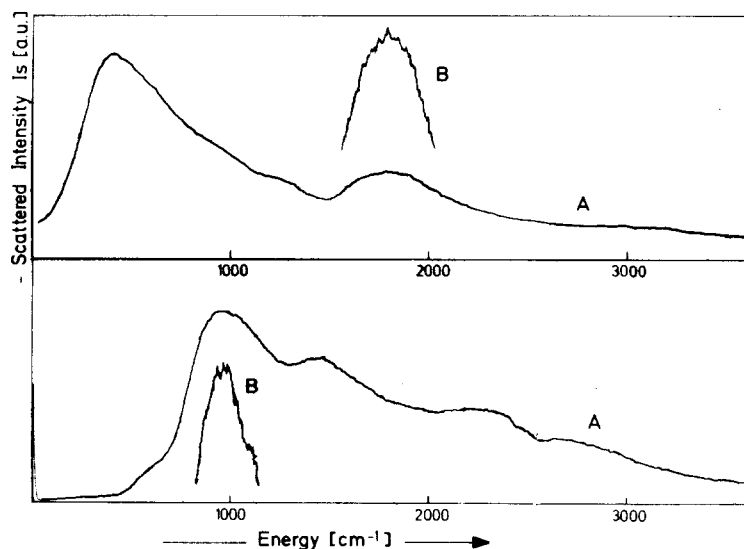


Fig. 6. Low temperature Raman shift spectra of *Escherichia coli* cell pastes obtained at 6 K. Top, laser at 5145 Å; Bottom, laser at 4880 Å. A = sensitivity at 100 K, B = sensitivity at 10 K.

plate. The technical details of the apparatus and procedures may be found in the publication by Drissler and Webb [27]. These spectra are reproduced here for comparison with the lines from time scans of active cells shown in figs. 3 and 4. The low temperature low sensitivity spectrum showed three distinct very broad bands while the high sensitivity scans revealed each of these bands to have a fine structure and thus the broad bands to be a result of the overlapping of many closely related Raman active modes. At low temperatures, this fine structure was completely reproducible. The interesting feature of the low temperature spectra was that the shift frequencies of broad humps appeared to correspond with the peaks in the average intensities and shift frequencies of the lines obtained with active cells (fig. 4). There was, however, one obvious difference in that many more Raman active modes were present in the fine structure of the low temperature humps than in the active cell spectra.

### Stokes spectra below $300\text{ cm}^{-1}$

At this stage of the studies little attention had been paid to shifts below  $300\text{ cm}^{-1}$  although it had been noticed that the Rayleigh wings, which were “smooth” with resting cells, always displayed more “noise” when active cells were investigated and, in some cases, definite sharp lines were seen superimposed on these wings. Moreover, the spectra obtained with non-synchronized cells displayed lines below  $200\text{ cm}^{-1}$  which were reproducible as far as shift frequencies were concerned but not intensity (fig. 7). These lines below  $200\text{ cm}^{-1}$  became of special interest because they did not occur in the published Raman spectra of solutions of isolated macromolecules surveyed by the author, but several groups had reported such lines in the spectra of crystals and other “solid” preparations of natural and synthetic proteins [28, 29, 30]. In our laboratory, moreover, such low frequency lines could not be detected in the spectra of concentrated, but aqueous, lysates of cells produced by bacteriophages or chloroform. It seemed, therefore, that the states associated with these lines were peculiar to the intact cell and crystals of biomolecules.

When synchronized cells were studied lines at low frequencies were seen in nearly every spectrum,

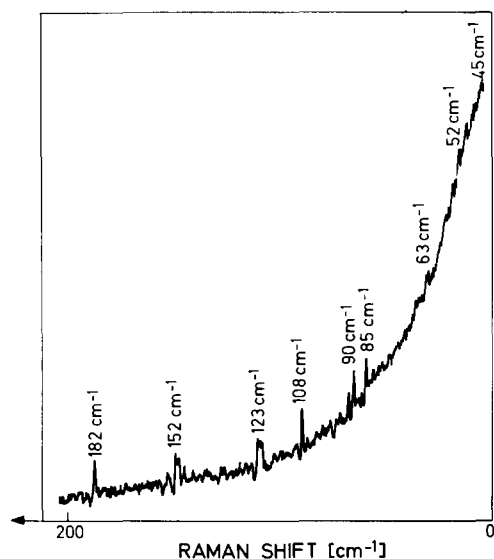


Fig. 7. The Raman shift spectrum, between 30 and  $200\text{ cm}^{-1}$  of non-synchronized cultures of *Escherichia coli* in Davis minimal medium. Cell concentrations  $5 \times 10^7/\text{ml}$ ; laser at  $5145\text{ \AA}$  and  $300\text{ mW}$ ; sensitivity  $10\text{ K}$ .

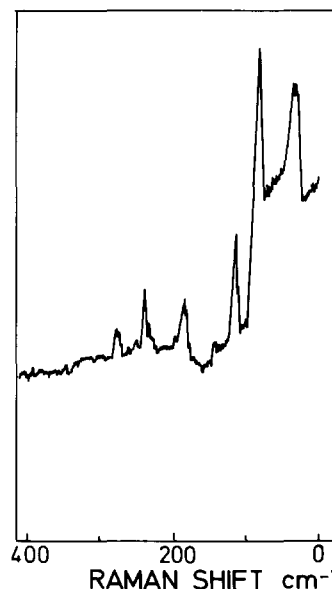


Fig. 8. A low frequency Raman shift spectrum obtained with cultures of *Escherichia coli* in which lines at  $45$  and  $85\text{ cm}^{-1}$  became very strong. Cell concentration  $2 \times 10^7/\text{ml}$ ; laser at  $5145\text{ \AA}$  and  $300\text{ mW}$ ; growth medium Davis minimal medium.

from  $0$  to  $200\text{ cm}^{-1}$ , taken 20 minutes after the start of cell incubation (i.e. 20 MAR). Their intensity varied, however, from appearing as a shoulder on the wings to very strong lines at  $45\text{ cm}^{-1}$ ,  $85\text{ cm}^{-1}$ ,  $150\text{ cm}^{-1}$  and  $186\text{ cm}^{-1}$  (figs. 8, 9). Such lines were seen in the spectra of both *Escherichia coli* and *B. megaterium* and in both cases their exact frequency and detection depended on the MAR time at which the spectrum was obtained. The line at about  $150\text{ cm}^{-1}$  appeared to move, as MAR time increased, from  $158\text{ cm}^{-1}$ , at the time of its first appearance, to  $126$  just before cell divisions began. The line at  $45\text{ cm}^{-1}$ , which was not detected until after 40 MAR, also moved, but only by a few wavenumbers, to  $38\text{ cm}^{-1}$  (see fig. 14). The suggestion that the lines moved came from consecutive scans from  $0$  to  $200\text{ cm}^{-1}$  each taking 80 seconds to run, in which only one predominant line between  $126\text{ cm}^{-1}$  and  $158\text{ cm}^{-1}$  was seen in any one scan (at least as far as the bacterial cells were concerned) which moved with time a few wavenumbers towards lower frequencies (fig. 10). Such lines at almost the same low wavenumbers have been observed recently in the spectrum of active algal cells and these also seem to change in intensity with time [32]. It is not known at present whether the apparent movement of the line is real or whether it is just another example of the time dependency of given Raman active states which, in this case, arise from a narrow cluster of *in vivo* states.

### Time scans

By now at least one thing had been established, this was that the Raman activity of *in vivo* states was a time-dependent phenomenon. It was natural, therefore, that time scanning should follow to ascertain

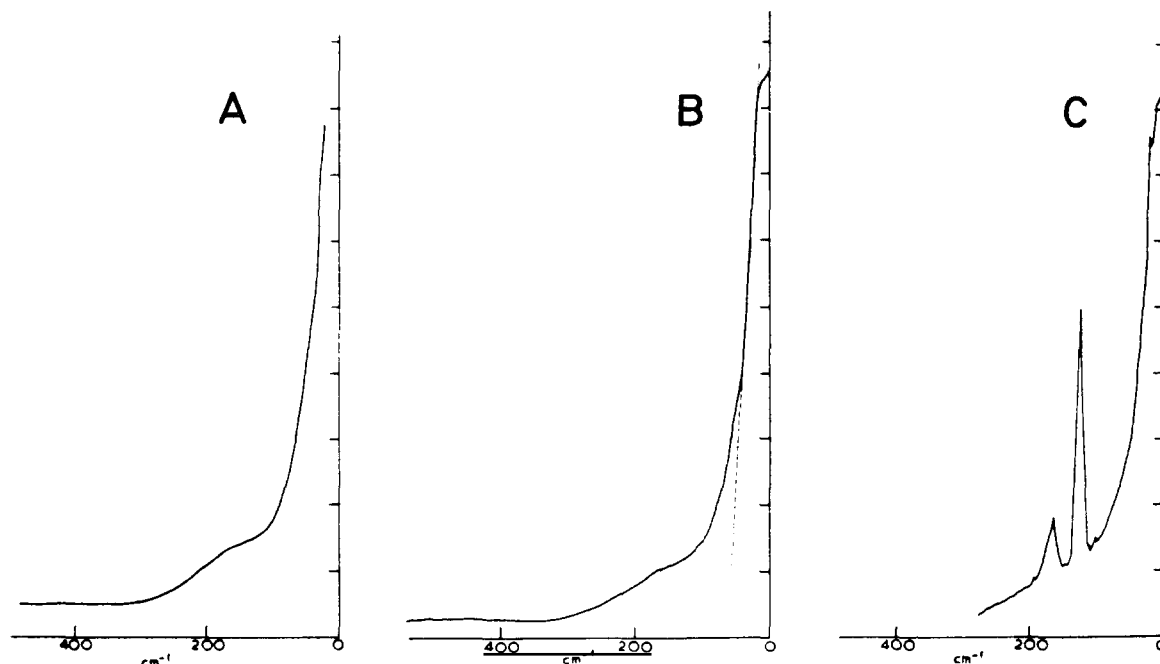


Fig. 9. Low frequency Raman spectra of A, Davis minimal medium; B, resting cells of *Bacillus megaterium*; C, active synchronized cells. Taken from Webb and Stoneham [31].

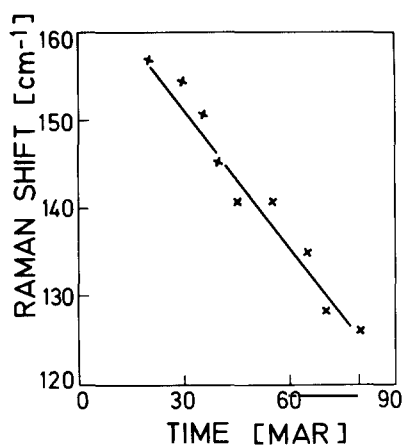


Fig. 10. The change seen in the frequency of a strong line between  $126\text{ cm}^{-1}$  and  $158\text{ cm}^{-1}$ , as cells of *Escherichia coli* progressed through a 90 minute life cycle.

the time, or times, during one cell cycle at which a given activity occurred. To do this a given shift frequency was chosen, the Raman instrument set on this frequency and a fresh sample of cells placed in the cuvettes every 5 minutes. The results repeated well the observations seen above (fig. 11) in that lines at the lower shift frequencies, but above  $300\text{ cm}^{-1}$ , were seen more abundantly early in the life of the cells and a gradual movement to higher frequency shifts took place as the life cycle of the cells progressed. Thus the time phenomenon would have been well established if it was not for a new finding. A given Raman activity at a given frequency certainly occurred at definite times in the life of the cell

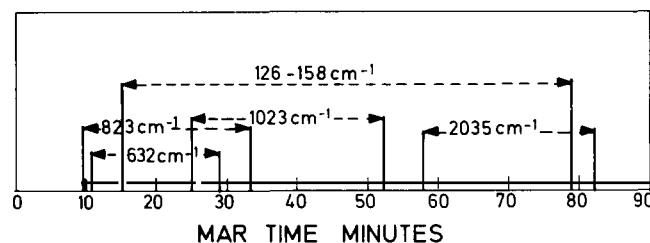


Fig. 11. The times at which set Raman shift lines occurred, in fixed shift frequency time scans, during a 90 minute cycle of synchronized cells of *Escherichia coli* in Davis minimal medium. Cell concentration  $3 \times 10^7$ /ml; laser 5145 Å and 300 mW.

but some of this activity appeared as “bursts” of lines over a 3 to 5 minute period (fig. 12). Since metabolic synchrony could be achieved only within 3–5 minutes a given line should have appeared at a set time and the line traced in amplitude, as a normal Gaussian curve indicating the normal distribution of the effective Raman active state within the 3–5 minute heterogeneity of the population. As seen in fig. 12 this, however, was not the case and so an analysis was made of the intervals of time between the appearance of successive strong lines ( $\Delta t$ ) on the spectral traces from many independent scans. The result, histogrammed in fig. 13, revealed that the interval between successive strong lines was not random;  $\Delta t$ 's of 15–20 sec and 50–55 sec apparently were the most commonly occurring intervals. The shaded portion of the histogram represents only the frequency with which the most common time interval occurred when all lines, including very weak ones, seen only in high resolution spectra, were included. It is shown, mainly because the computer analysis revealed its presence but, since many of the lines analysed were extremely weak, it must be viewed with scepticism. The  $\Delta t$ 's of the strong lines, which are well under the minimal 3 minute synchrony time, however, seem real and this phenomenon is discussed below.

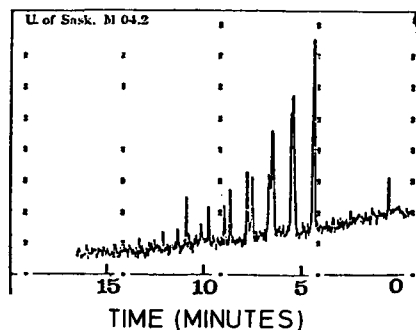


Fig. 12. Bursts of Raman activity which occurred at  $1273 \text{ cm}^{-1}$  over a 5 minute period and seen in fixed frequency time scans of synchronized cultures of *Escherichia coli*. From Stoneham and Webb [33].

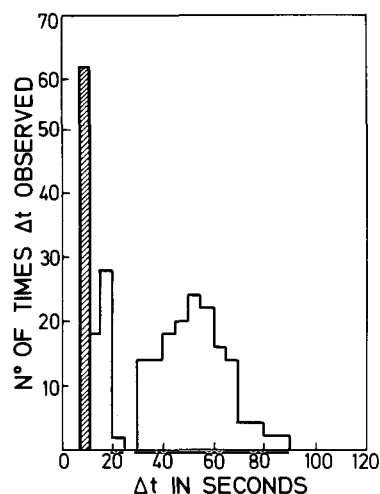


Fig. 13. Histogramme of the intervals of time ( $\Delta t$ ) which occurred between the sequential appearance of a line at  $1273 \text{ cm}^{-1}$  within a 3 to 5 minute period during fixed frequency time scans with synchronized *Escherichia coli* cells.

## Anti-Stokes lines

At this stage of the investigations the anti-Stokes spectra had not been explored since it was assumed that the intensity of such lines above  $300\text{ cm}^{-1}$  would be too low to be detected. However, the appearance of Stokes lines below  $200\text{ cm}^{-1}$  which, at times were of considerable strength, made it reasonable to search for them. Such lines were observed but again their intensity varied from run to run and many appeared as broad bands. At the MAR time when the Stokes line reached its maximum intensity so did the corresponding anti-Stokes line but detailed analysis of their intensity ratio proved difficult because of the need to change the cell sample between successive spectral runs. This required many cultures of cells each started (0 MAR) at various times so that each population could be examined at the same MAR time. The inherent small, but uncontrollable, variation in the time of the cell cycle, from culture to culture, made it impossible to obtain the same absolute intensity of the lines. One of the best sequences observed is shown in fig. 14, but it must be emphasized that this has been selected from many spectra of which most were not of the same quality. To obtain this type of spectrum the sensitivity of the instrument was set at  $1 \times 10^5$  cps during the first  $30\text{ cm}^{-1}$  wavenumbers of the scan past the laser line and then switched through  $1 \times 10^4$  to  $5 \times 10^3$  cps. This procedure tended to flatten the wings, result in small "false" lines over the first  $35\text{ cm}^{-1}$  and amplify the intensity of real lines between  $40$  and  $60\text{ cm}^{-1}$  which appear on the Rayleigh wings. The line at about  $150\text{ cm}^{-1}$ , therefore, is a true representation of its Stokes and anti-Stokes intensities, that at  $43\text{ cm}^{-1}$ , however, must be viewed with caution. This caution kept in mind, it is interesting nevertheless that both Stokes and anti-Stokes at  $43\text{ cm}^{-1}$ , have almost the same intensity. This type of intensity ratio also held true of the line at  $158\text{ cm}^{-1}$  which, as stated above, appeared to move to lower wavenumbers as the MAR time increased. In eight successive spectral runs the anti-Stokes to Stokes ratio of this line was found to vary from a low of 0.73 to a high of 1.08.

As was done with the Stokes lines some time runs were conducted at set anti-Stokes shift frequencies

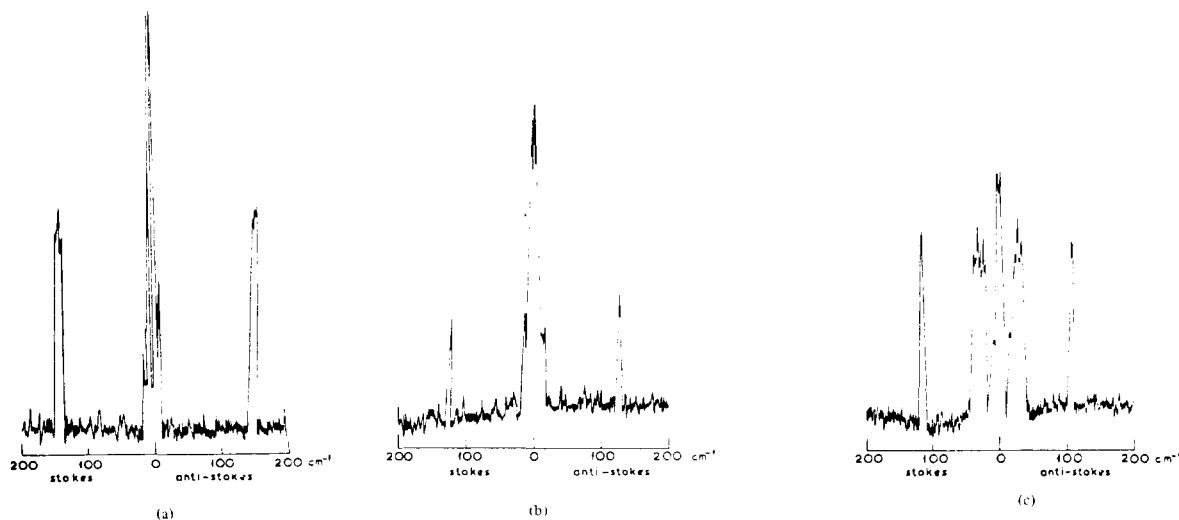


Fig. 14. An example of Stokes and anti-Stokes low frequency spectra of *Escherichia coli* cells obtained at 40, 50 and 60 minutes after resuspension. Taken from Webb et al. [34].

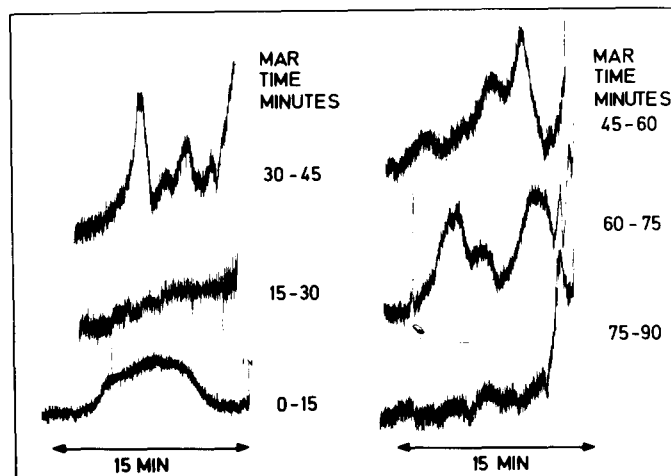


Fig. 15. The changing pattern of anti-Stokes line at  $455\text{ cm}^{-1}$  seen during fixed frequency time scans with synchronized cells of *Escherichia coli* MAR TIME = minutes after resuspension over which a 15 minute time scan was conducted. Laser at  $5145\text{ Å}$ .

and one line at  $455\text{ cm}^{-1}$  so far has been studied. This line was selected because of its relatively low wavenumber and its frequent appearance in normal spectral scans. As can be seen from fig. 15 the characteristics of the line altered considerably with MAR time, also the line did not reappear if a given suspension of cells was left unchanged in the laser light for more than 10–15 minutes, if, at this time, however, a new sample of cells was introduced the line reappeared during the next 15 minute period. Thus this anti-Stokes line behaved in a similar manner to that of many Stokes lines in that its intensity and appearance fluctuated with MAR time and the *in vivo* activity, responsible for the Raman activity, seemingly sensitive to some “bleaching” action of the laser light, if exposure to the light was prolonged. Bleaching was indicated because the length of time, in the laser light, over which the line could be seen, appeared to be inversely proportional to the laser power used.

### Cells other than bacteria

In recent times both Stokes and anti-Stokes lines at relatively high shift frequencies have been observed in the spectrum of algal cells exposed for only a millisecond or so to light to activate photosynthesis. The lines, seen at 300 K, however, were those of the carotenoid pigments only, the spectral lines associated with the large chlorophyll molecules were not detected (fig. 16). Chlorophyll lines, however, now have been seen clearly in the spectrum of these algal cells held at low temperatures (fig. 17) and, moreover, in experiments where the temperature has been gradually increased from 6 K to 300 K, these lines have been found to “disappear” from the spectrum at about 273 K, i.e. at the melting point of water. In addition, at 300 K and certain laser frequencies, lines below  $200\text{ cm}^{-1}$  which vary greatly in intensity from run to run and with laser wavelength have been observed. A detailed description of these thermal effects will appear shortly in a new publication [37]. The lack of appearance of the chlorophyll lines and indeed other lines, seems to simulate the non appearance of lines in the spectra of resting bacterial cells at 300 K, however, it is possible that some chlorophyll and carotenoid lines overlap.

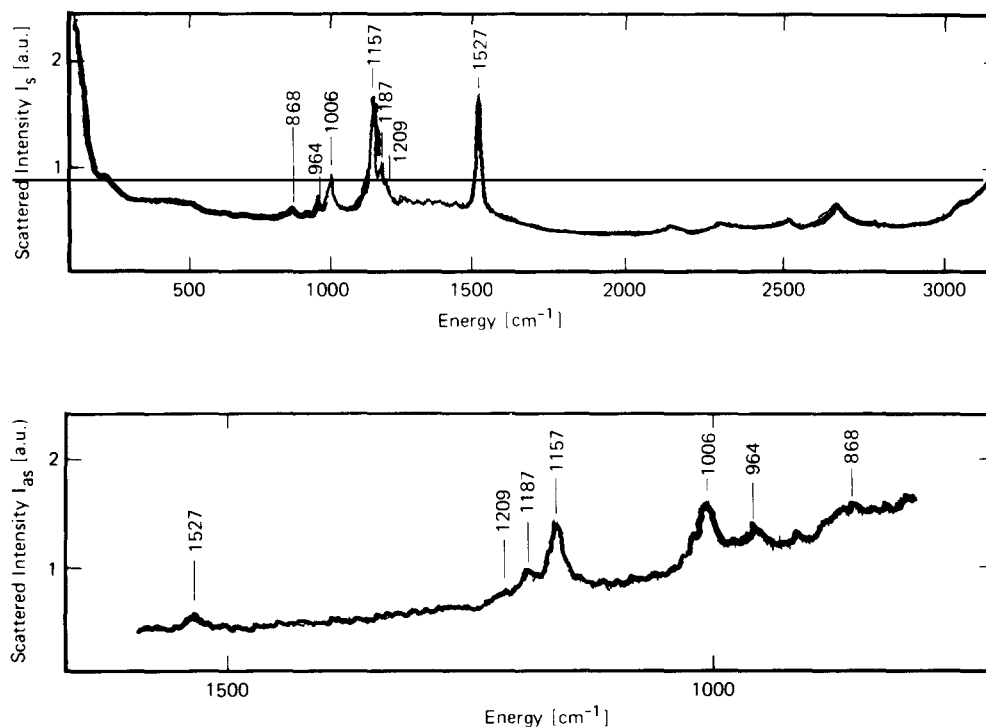


Fig. 16. Raman shift spectra of *Chlorella pyrenoidosa*. Top, Stokes lines; Bottom, anti-Stokes lines. Taken from Drissler and MacFarlane [35]. The Stokes bands shown, have been assigned to vibrational modes of carotenoid molecules by Lutz [34].

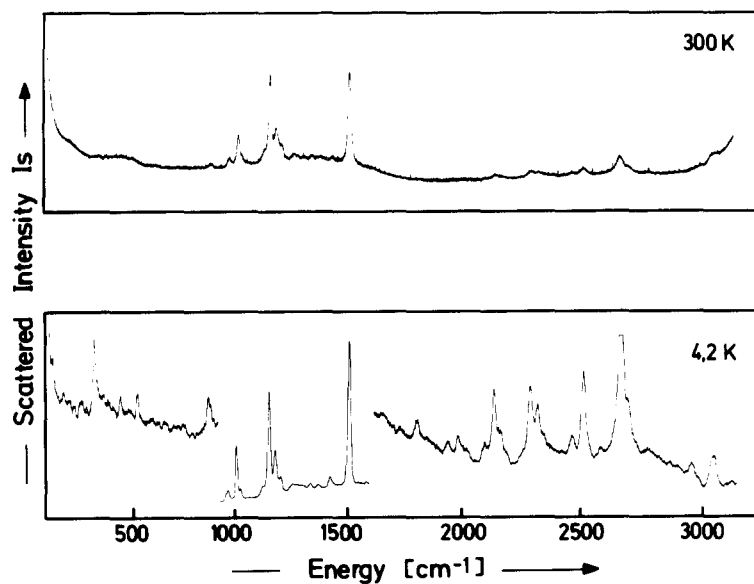


Fig. 17. Stokes spectra of *Chlorella pyrenoidosa* at 300 K (top) and at 4.2 K (bottom). Courtesy of F. Drissler [37].

## Normal and tumour animal cells

In fig. 18 are shown some recently determined spectra of "normal" and tumour mammary cells, between 0 and  $200\text{ cm}^{-1}$ . These were obtained by taking scrapings from the outer regions of freshly removed 2 to 3 diameter mammary carcinomas or pieces of normal tissue [38]. The scrapings, containing pieces of tissue about 1.0 mm in thickness, were washed in Dulbecco's salts solution and then placed in a tissue culture medium similar to that of the commercial 1066 medium. Modifications, however, had to be made in this medium in that dyes and animal serum were omitted because, if present, both produced fluorescences which tended to hinder the characteristics of the spectra obtained. Such cells, if suspended in holding fluids such as Hanks (without dyes) and Dulbecco's basic salt solutions, produced no discernable Raman activity between 0 and  $200\text{ cm}^{-1}$  nor did those placed in a tissue culture medium several hours after they had been removed from a patient or subjected previously to "fixing" solutions, only with the fresh tissue was the Raman activity, shown in fig. 17, seen. An interesting feature of these spectra was that the Raman lines from tumour tissue and the "normal" tissue from a diseased patient were broad and double spiked areas whereas the spectral lines from normal tissue, i.e. from a patient found not to be suffering mammary carcinoma, were narrower and sharp. Further studies of this phenomenon seem warranted, therefore, to ascertain whether the pattern of Raman activity seen is characteristic of other tumour cells.

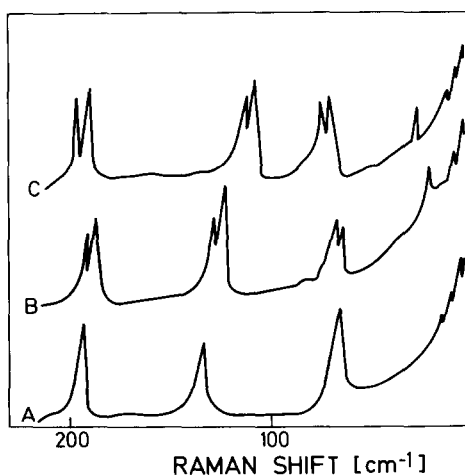


Fig. 18. Low frequency Raman shift spectra of: A, normal mammary tissue; B, "normal" right side tissue; C, left side 2–3 cm diameter mammary carcinoma of a human female. Taken from Webb et al. [38].

## Discussion

The studies presented paint a complex picture and pose more questions than answers, a great deal more work will be required, therefore, before any sound conclusions may be made as to what the above findings mean. With this caution in mind, it does appear that at physiological temperatures the resting cell is in a stable condition with few, if any, of its modes Raman active. This stability is thought to be due to the presence and activity of those water molecules which form an intricate lattice *in vivo* and around macromolecules and their complexes. It may be argued that the Raman lines are not seen in the

spectra of resting cells in suspension is because, at a concentration of  $10^7$  cells/ml, their total volume is too small. If this was the case, however, no lines would be detected in the spectra of any such suspension but, as demonstrated, the active cell spectra display many lines which often have a very high intensity. If the fine low temperature spectra of *Escherichia coli* is compared with the composite active cell spectra it becomes evident that fewer lines within a given  $300$  to  $400\text{ cm}^{-1}$  band, appear in the active spectra than in the low temperature ones, but because of the small volume of cells used in the former case, the lines in the active spectra are very much stronger. The impression is given then that some form of condensation of closely related states takes place as a result of metabolic activity which gives rise to Raman activity, at a given frequency, of a high amplitude. Thus the Stokes lines in the active cell spectra may arise from a condensation *in vivo* of states excited by metabolic processes as suggested by Fröhlich [5, 6].

An indication that a condensation type of process does take place comes from the ratios of respective Stokes and anti-Stokes lines, and indeed the appearance of anti-Stokes lines above  $500\text{ cm}^{-1}$ . Under normal thermal equilibrium conditions the intensity of anti-Stokes lines above  $500\text{ cm}^{-1}$  would be too small to be detected while the intensity of those below  $200\text{ cm}^{-1}$  would be approximately half that of their Stokes counterpart. As pointed out by Drissler and MacFarlane [35], the intensity ratio  $R$  at laser frequency  $\omega$  is given by:

$$R = \frac{I_{as}^{\omega}}{I_s^{\omega}} = A \frac{(\omega - \omega_e - \Omega)^2}{(\omega - \omega_e + \Omega)^2} P[\Omega, T],$$

where  $A$  accounts for reabsorption, the second factor, the resonance term (with  $\Omega$  the vibrational mode frequency) and  $P$  the thermal population ratio ( $n/(n+1)$ ) derived from Boltzmann's statistics in which  $n$  is the number of vibrators in a particular energy level.

With cells containing a large quantity of molecules which have a strong absorption in the visible at about  $480\text{ nm}$ , such as the carotenoids, the anti-Stokes line is nearer resonance with these pigments than is the Stokes line when the laser is set at say  $514.5\text{ nm}$ . Near resonance, and in particular for large  $\Omega$ ,  $R$  could be influenced, and perhaps dominated, by preferential resonance enhancement in the intensity of the anti-Stokes line due to resonance with the carotenoid absorption band at  $480\text{ nm}$ ; i.e., mathematically, by the cross sectional factor. In recognition of this the authors [35] attempted to separate resonance factors from the population ratio  $P$  by a consideration of the ratio between the Stokes intensity  $I_s^{\omega+\Omega}$  (laser at  $\omega + \Omega$ ) and the anti-Stokes intensity  $I_{as}^{\omega}$  (laser at  $\omega$ ) because under these conditions

$$P = I_{as}^{\omega} / I_s^{\omega+\Omega},$$

from which

$$P = \frac{I_{as}^{\omega}}{I_s^{\omega+\Omega}} \cdot \frac{I_s^{\omega}}{I_s^{\omega}} = \frac{I_{as}^{\omega}}{I_s^{\omega}} \cdot \frac{I_s^{\omega}}{I_s^{\omega+\Omega}} = R \frac{I_s^{\omega}}{I_s^{\omega+\Omega}}.$$

Thus  $R$  was measured at a given  $\omega$  and corrected from a plot of the intensities of the Stokes lines at  $868\text{ cm}^{-1}$ ,  $1157\text{ cm}^{-1}$ ,  $1006\text{ cm}^{-1}$  and  $1527\text{ cm}^{-1}$  found with equal laser power but different wavelengths (fig. 19). With their flow system, in which the cells were exposed to light for only a millisecond or so, they calculated, under these conditions, for the  $4\Omega$ 's, population enhancements, over the thermal population factor, of 1.2 to 2.6. They found, however, in preliminary experiments, that, if the cells were exposed to white light for longer periods, the intensity of the Raman lines was increased by approximately 5%; this increase, if proven true would yield a much greater population enhancement

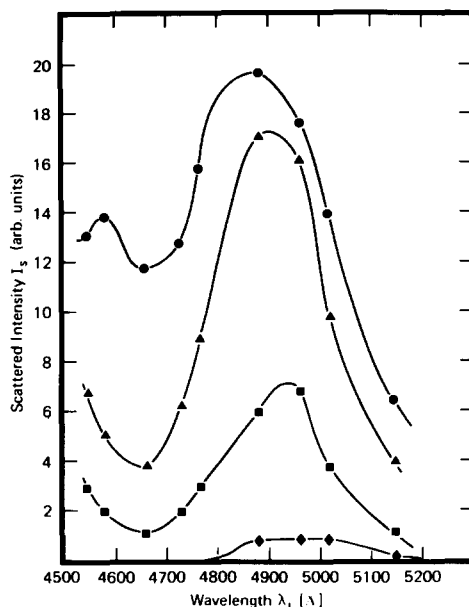


Fig. 19. Intensity dependence of the Stokes shifted bands at  $868\text{ cm}^{-1}$  (♦),  $1006\text{ cm}^{-1}$  (■),  $1157\text{ cm}^{-1}$  (▲) and  $1527\text{ cm}^{-1}$  (●) on the wavelength of the incident laser light. Taken from [35].

over the thermal factor. As far as the bacteria studied are concerned the anti-Stokes/Stokes ratio ( $R$ ) at  $150\text{ cm}^{-1}$  had an average value of about 0.8 which, at this frequency, indicates a very large population enhancement over the thermal factor. It is not known, at present, whether resonance enhancements of the above type occur in bacteria able to influence  $R$  in the same way. At this time it seems unlikely because they do not possess, in great quantities, the carotenoid pigments present in algal cells and display no absorption in the visible but further work with laser wavelengths other than the  $5145\text{ Å}$  and  $4880\text{ Å}$  wavelengths used will clarify this point. In any event, in two independent studies the anti-Stokes/Stokes ratios have been found to exceed considerably those predicted by thermal population factors. Thus metabolic processes do appear to excite, and perhaps produce, by non-thermal means specific Raman active modes.

One thing that certainly emerges from the studies presented is the time dependency of the onset of *in vivo* Raman activity at specific frequencies. This suggests that the living cell only allows small “cracks” to occur in its otherwise Raman inactive stable state and somehow controls, in time and frequency, the occurrence of these cracks. It is assumed, therefore, that the great stability and accuracy of function of the living system, which is in a constant state of flux, resides in its ability to confine the occurrence of unstable states, which are required for metabolic activities to take place, to set times in its lifetime. In this way it maintains its dynamic stability by allowing only minute “cracks”, or perhaps one crack, to occur at any given time while the rest of the system is held in a stable state. If, as suspected, water molecules are involved in the production of the stable system, then since nearly all *in vivo* metabolic processes involve the addition or subtraction of one or two  $\text{H}_2\text{O}$  molecules to a given molecule, the movement of  $\text{H}_2\text{O}$  at a specific time and place could lead to the “cracks” in the otherwise stable architecture of the cell and lead to Raman activity of the groups with which  $\text{H}_2\text{O}$  directly associates or dissociates. It has been suggested that the known timing and occurrence of events *in vivo* may arise from an overall uniform spiral, or helical motion within the cell which brings together enzymes and their

substrates at specific times. The pitch of this motion, determined by nutritional factors, thus controls when, and if, an event will take place [39, 40, 41, 42]. In recent preliminary studies nutritional factors, both in terms of the type of carbon and nitrogen supplied to the cells, have been found to alter the frequencies and timing of Raman active events but it will take much effort to establish just how each nutritional factor alters the *in vivo* "Raman active time clock".

The Raman lines below  $200\text{ cm}^{-1}$  clearly show a different behaviour in time from those above  $200\text{ cm}^{-1}$ . First, these lines are present in the spectra over a prolonged period of the cells' life time, even though they may vary in intensity from a mere shoulder on the Rayleigh wings to the strong lines shown in figs. 8, 9, 14 and second, their movement with time appears to be towards lower frequencies. The metabolic process associated with these Raman lines, therefore, seems to be of a different nature from those producing lines at higher shift frequencies. Such low frequency Raman lines have been seen only in the spectra of large crystals of proteins and then it has required special techniques for their presence to be detected. In the living cell enzyme molecules of a given type are known to be clustered together in blocks, thus the lines below  $200\text{ cm}^{-1}$  may arise from collective coherent modes peculiar to blocks of proteins, or the cell membrane, which, with time, increase in size as their metabolic synthesis proceeds.

In this report no effort has been made to identify any of the Raman lines with specific metabolic activities, or indeed the particular modes of biosystems. Only in one case has this been attempted by blocking the respiratory chain with carbon monoxide which resulted in severe changes in the characteristics, with time, of a line at  $1273\text{ cm}^{-1}$  [33]. To identify each of the lines by the specific inhibition of set metabolic processes obviously will require an enormous amount of work, if the techniques of analysis of many time spectra, used in this study, is employed. What is required is instrumentation capable of displaying simultaneously all the Raman lines present at a given MAR time; such equipment is available but expensive. In addition, there is the danger that the inhibition of one *in vivo* process may affect the behaviour of others, a factor that could lead to falsely identifying a line with a given process. Some method should be found, however, to achieve this goal and perhaps the best one would be to study, as mentioned above, the effects of various types of nutrients on the actual shift frequencies seen and their time of appearance. Much is known about the biochemical and molecular changes induced *in vivo* by nutritional factor and it is possible that this knowledge could be put to use to identify lines.

The "bursts" of lines, seen at set MAR times but occurring within the 3–5 minute synchrony, is a mystery. This pattern could indicate that some sort of metabolic pumping, of modes with frequencies close to the Raman line, occurs over a few seconds which produces a coherent mode of high amplitude. It is difficult to comprehend, however, how this could be seen in the spectra of populations of cells metabolically, "out of phase" by at least 3 minutes. A second possibility is that some sort of resonance enhancement, induced by the laser at the frequencies the phenomenon is seen, takes place and that alternate laser excitation and biologically induced energy transfer produce this type of pattern; perhaps future work will shed some light on the phenomenon.

Time resolved Raman spectroscopy, of time scales from  $10^{-1}$  to  $10^{-8}$  seconds has been achieved with isolated bacteriorhodopsin, a retinal-proteins complex found in some halophilic bacterial cells [43–46]. Such studies have shown changes in the spectrum of the complex within this small timescale, and it may be that the phenomenon has some relationship to "bursts" of lines seen in this study, however, the time-scale here is at least one order of magnitude higher.

The very sharp "spikes" with narrow bandwidths, sometimes seen on the tops of weak broad bands, are thought to be monochromatic fluorescences induced by the laser. The emission of virtually monochromatic light by some genera of bacteria and fungi as a result of metabolic activity, is not uncommon. In the present case, however, if the laser light was removed, no such spikes were seen; so they do not appear to arise from the metabolic activities of the cells used in this study.

One other fact that seems to arise from the study presented is that the living cell system appears unique in its ability to control the occurrence of Raman active states. The closest entities to life forms are the viruses and independent studies of aqueous suspension of bacterial [47] and plant [48, 49, 50] viruses have shown them to have a clear reproducible spectra, between  $200\text{ cm}^{-1}$  and  $3100\text{ cm}^{-1}$  which possess lines of high intensity. These entities are macromolecular complexes of protein and nucleic acid, and much smaller than bacterial cells. The question arises, therefore, as to why viruses display such high Raman activity while resting bacterial cells, in suspension, do not. Locked up in this phenomenon, therefore, may be a key as to why and how viruses are able to infect cells, disrupt and take control of their metabolic activities. Here again studies of the effect of virus infection on the time dependent Raman spectra of cells should prove enlightening and may provide a new tool by which virus infection may be better understood from the standpoint of the physics of the process.

Finally the phenomenon shown as far as the Raman spectra of normal versus tumour cells are concerned [26, 51]. In the one clinical case shown [38], there is a clear difference in the characteristics of the Raman lines below  $200\text{ cm}^{-1}$ . It is possible, therefore, that in normal cells degenerate states exist which, as a result of the oncogenic process, becomes resolved and this process produces the broad double spiked peaks seen in the tumour cell spectrum. Should this prove true of other tumour cells, then Raman spectroscopy, and perhaps time dependent spectroscopy, may prove a new method by which tumour, and pretumour, cells may be rapidly identified. Indeed, differences in the Raman spectra of the membranes of such cells already have been reported [52].

The study described, at this stage, has solved very little and has merely confirmed facts cell biologists have uncovered by other means, concerning the existence of an accurate metabolic time clock within living cells. It has introduced nevertheless a different parameter in the form of the physical nature of this clock and a new method by which it may be studied. Clearly, however, the above discussion concerning "why Raman spectroscopy" is grossly inadequate as groundwork for any interpretation to be made here as to the seemingly complex physics governing the observations shown. It is hoped, therefore, that these findings will encourage some theoretical physicists to explore, in detail, their possible meanings. In addition the work seems to have opened a new approach by which a variety of different biological phenomena, such as nutrition, virus infection and oncogenesis may be identified and studied with respect to energetics and time. There is, therefore, much to be done.

## Acknowledgements

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